Supplementary Material for "Multi-Horizon Failure-Risk Models for Battery Failure Prediction: How Much Cross-Chemistry Transfer Is a State-of-Health Shortcut?"

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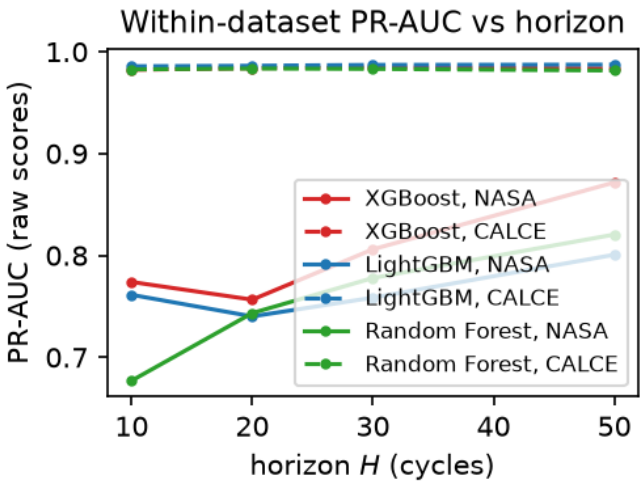


Fig. 1. Within-dataset PR-AUC of raw scores versus horizon (main manuscript, Section III-B). Precision–recall performance, unlike ROC-AUC, is sensitive to the horizon-dependent failure prevalence; it is also where isotonic's tied step outputs pay for their ranking-free compression.

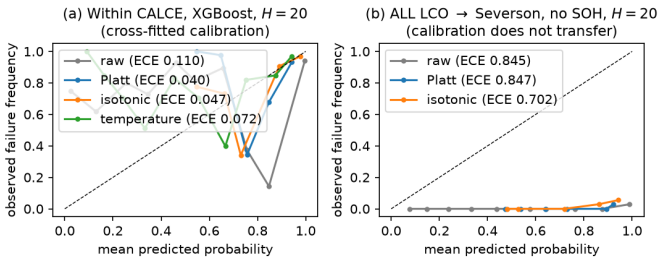


Fig. 2. Reliability diagrams under the cross-fitted protocol. (a) Within CALCE, XGBoost at H=20: the fitted Platt sigmoid preserves ranking but washes confidence toward the base rate. (b) ALL-LCO → Severson without SOH at H=20: no fitted calibrator, whichever source scores it is applied to, repairs calibration on the shifted target.

Contents

This supplement contains the tables and figures omitted from the main manuscript for length: the diagnosis-versus-prognosis split (Table I), the factorial feature ablation (Table II), GRU and hazard-model transfer (Tables III and IV), monotonicity violations (Table V), the full SOH-ablation statistical evidence (Table VI), the recalibration arms (Tables VII and VIII), the leave-condition-out analysis (Table IX), and eleven supplementary figures. Section references of the form "Section III-x" point to the main manuscript.

S-I. Supplementary Tables
S-II. Supplementary Figures

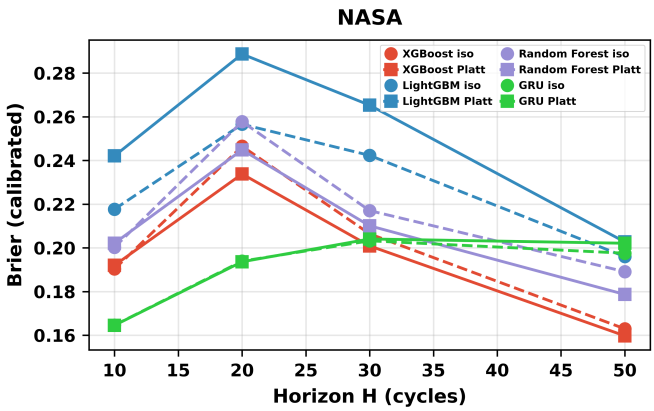


Fig. 3. Calibration curves on NASA at H=20: raw, Platt, isotonic, and temperature-scaled scores against the observed failure frequency.

TABLE I

DIAGNOSIS VERSUS PROGNOSIS AT H=20 (ALL-LCO SOURCE, RAW SCORES): ALL-ROWS POOLED AUC [CELL-LEVEL CI WHERE COMPUTED] AGAINST PROGNOSIS-ONLY AUC ON ROWS SCORED WHILE SOH>0.80, WITH THE SHARE OF ROWS SCORED ABOVE THRESHOLD. GRU AND DISTANCE-RULE ROWS ARE POINT ESTIMATES.

Target	Model	Features	All rows	Prognosis-only	Frac. above
Severson	GRU	common, no SOH	0.631	0.553	0.99
Severson	GRU	common+SOH	0.827	0.756	0.99
Severson	LightGBM	no SOH	0.718 [0.634,0.846]	0.674 [0.597,0.851]	0.99
Severson	LightGBM	with SOH	0.882 [0.786,0.990]	0.814 [0.732,0.978]	0.99
Severson	SOH-dist rule	dist rule	0.912	0.862	0.99
Severson	SOH-dist rule	SOH-only	0.912	0.862	0.99
Severson	Random Forest	no SOH	0.746 [0.673,0.845]	0.705 [0.632,0.843]	0.99
Severson	Random Forest	with SOH	0.889 [0.783,0.998]	0.825 [0.737,0.996]	0.99
Severson	XGBoost	no SOH	0.750 [0.692,0.835]	0.721 [0.666,0.833]	0.99
Severson	XGBoost	with SOH	0.896 [0.804,0.995]	0.838 [0.760,0.991]	0.99
HNEI NMC	LightGBM	no SOH	0.988 [0.974,0.996]	0.905 [0.863,0.944]	0.28
HNEI NMC	LightGBM	with SOH	0.999 [0.998,0.999]	0.969 [0.946,0.987]	0.28
HNEI NMC	Random Forest	no SOH	0.996 [0.994,0.999]	0.945 [0.908,0.978]	0.28
HNEI NMC	Random Forest	with SOH	0.999 [0.999,1.000]	0.981 [0.970,0.993]	0.28
HNEI NMC	XGBoost	no SOH	0.989 [0.977,0.997]	0.908 [0.870,0.945]	0.28
HNEI NMC	XGBoost	with SOH	0.993 [0.983,0.999]	0.963 [0.935,0.987]	0.28
SNL NCA	LightGBM	no SOH	0.850 [0.791,0.911]	0.841 [0.798,0.889]	0.63
SNL NCA	LightGBM	with SOH	0.985 [0.974,0.993]	0.891 [0.845,0.946]	0.63
SNL NCA	Random Forest	no SOH	0.911 [0.875,0.951]	0.837 [0.779,0.907]	0.63
SNL NCA	Random Forest	with SOH	0.983 [0.974,0.992]	0.901 [0.858,0.957]	0.63
SNL NCA	XGBoost	no SOH	0.878 [0.824,0.930]	0.849 [0.803,0.895]	0.63
SNL NCA	XGBoost	with SOH	0.972 [0.956,0.988]	0.886 [0.837,0.943]	0.63
SNL NMC	LightGBM	no SOH	0.693 [0.570,0.959]	0.728 [0.577,0.956]	0.60
SNL NMC	LightGBM	with SOH	0.847 [0.743,0.982]	0.722 [0.604,0.956]	0.60
SNL NMC	Random Forest	no SOH	0.742 [0.689,0.874]	0.649 [0.604,0.793]	0.60
SNL NMC	Random Forest	with SOH	0.858 [0.748,0.975]	0.706 [0.613,0.929]	0.60
SNL NMC	XGBoost	no SOH	0.709 [0.612,0.953]	0.730 [0.599,0.946]	0.60
SNL NMC	XGBoost	with SOH	0.772 [0.677,0.979]	0.729 [0.608,0.960]	0.60
SNL LFP	LightGBM	no SOH	0.359 [0.282,0.443]	0.383 [0.311,0.467]	0.97
SNL LFP	LightGBM	with SOH	0.999 [0.998,1.000]	0.993 [0.988,0.998]	0.97
SNL LFP	Random Forest	no SOH	0.666 [0.569,0.769]	0.588 [0.565,0.616]	0.97
SNL LFP	Random Forest	with SOH	0.999 [0.998,1.000]	0.994 [0.990,0.999]	0.97
SNL LFP	XGBoost	no SOH	0.496 [0.455,0.531]	0.517 [0.481,0.575]	0.97
SNL LFP	XGBoost	with SOH	0.998 [0.997,1.000]	0.991 [0.985,0.997]	0.97

TABLE II

FACTORIAL FEATURE ABLATION AT H=20 (XGBOOST, POOLED AUC WITH CELL-LEVEL BOOTSTRAP 95% CI). FEATURE GROUPS: SOH; CYCLE INDEX; SENSORS = VOLTAGES, CURRENT, TEMPERATURE, DURATION.

	Oxford			Severson		
Feature set	**NASA src**	**CALCE src**	**ALL src**	**NASA src**	**CALCE src**	**ALL src**
full	0.957 [0.931,0.987]	0.850 [0.764,0.910]	0.917 [0.878,0.965]	0.871 [0.767,0.997]	0.850 [0.749,0.967]	0.896 [0.812,0.995]
no SOH	0.518 [0.513,0.522]	0.416 [0.264,0.554]	0.429 [0.279,0.597]	0.595 [0.510,0.726]	0.682 [0.670,0.695]	0.750 [0.692,0.835]
−cycle	0.965 [0.942,0.989]	0.967 [0.929,0.981]	0.993 [0.954,1.000]	0.854 [0.733,0.998]	0.766 [0.584,0.974]	0.774 [0.578,0.998]
−SOH−cycle	0.658 [0.522,0.797]	0.433 [0.282,0.594]	0.388 [0.224,0.506]	0.531 [0.433,0.680]	0.500 [0.500,0.500]	0.535 [0.455,0.653]
SOH only	0.965 [0.942,0.989]	1.000 [1.000,1.000]	1.000 [1.000,1.000]	0.834 [0.695,0.991]	0.798 [0.638,0.978]	0.833 [0.689,0.993]
cycle only	0.541 [0.537,0.545]	0.541 [0.537,0.545]	0.541 [0.537,0.545]	0.709 [0.695,0.724]	0.694 [0.681,0.708]	0.719 [0.704,0.735]
sensors	0.658 [0.522,0.797]	0.433 [0.282,0.594]	0.388 [0.224,0.506]	0.531 [0.433,0.680]	0.500 [0.500,0.500]	0.535 [0.455,0.653]

TABLE III

GRU TRANSFER AT H=20 ON THE COMMON FEATURE SET: SEED-AVERAGED PER-CELL AUC (± SD OVER SEEDS 42, 1, 7).

Training	Feature set	Oxford	Severson
NASA	common w/ SOH	0.196±0.155	0.998±0.003
NASA	common no SOH	0.147±0.210	0.987±0.015
CALCE	common w/ SOH	0.182±0.311	0.964±0.033
CALCE	common no SOH	0.099±0.171	0.479±0.451
ALL LCO	common w/ SOH	0.088±0.084	0.991±0.015
ALL LCO	common no SOH	0.000±0.000	0.893±0.131

TABLE IV

DISCRETE-TIME HAZARD MODEL AT H=20: TRANSFER PERFORMANCE (POOLED AUC [CI]). THE FIXED-HORIZON XGBOOST COMPARATOR WITH SOH IS IN TABLE V OF THE MAIN MANUSCRIPT.

Training	Model	Oxford	Severson
NASA	Hazard XGBoost	0.971 [0.953,0.991]	0.884 [0.786,0.998]
NASA	Hazard Logistic	0.500 [0.500,0.500]	0.607 [0.402,0.890]
CALCE	Hazard XGBoost	0.466 [0.301,0.578]	0.737 [0.711,0.774]
CALCE	Hazard Logistic	0.500 [0.500,0.500]	0.500 [0.500,0.500]
ALL LCO	Hazard XGBoost	0.314 [0.135,0.457]	0.866 [0.770,0.978]
ALL LCO	Hazard Logistic	0.500 [0.500,0.500]	0.916 [0.873,0.969]

TABLE V
MONOTONICITY VIOLATIONS: SHARE OF SCORED ROWS WHERE THE CUMULATIVE-RISK ORDERING FAILS ON RAW SCORES (PLATT IN PARENTHESES). THE SURVIVAL PRODUCT IS MONOTONE BY CONSTRUCTION (RATE 0).

Setting	Target	Model	Raw	Platt
within	calce	XGBoost	0.452	0.056
within	nasa	XGBoost	0.323	0.260
transfer (calce src)	oxford	LightGBM	0.641	0.004
transfer (calce src)	oxford	Random Forest	0.501	0.540
transfer (calce src)	oxford	XGBoost	0.565	0.003
transfer (nasa+calce src)	oxford	LightGBM	0.210	0.648
transfer (nasa+calce src)	oxford	Random Forest	0.335	0.305
transfer (nasa+calce src)	oxford	XGBoost	0.609	0.341
transfer (nasa src)	oxford	LightGBM	0.521	0.468
transfer (nasa src)	oxford	Random Forest	0.559	0.376
transfer (nasa src)	oxford	XGBoost	0.248	0.124
transfer (calce src)	severson	LightGBM	0.405	0.013
transfer (calce src)	severson	XGBoost	0.420	0.072
transfer (nasa src)	severson	LightGBM	0.582	0.537
transfer (nasa src)	severson	Random Forest	0.778	0.557
transfer (nasa src)	severson	XGBoost	0.958	0.923
—	—	Survival product	0.000	0.000

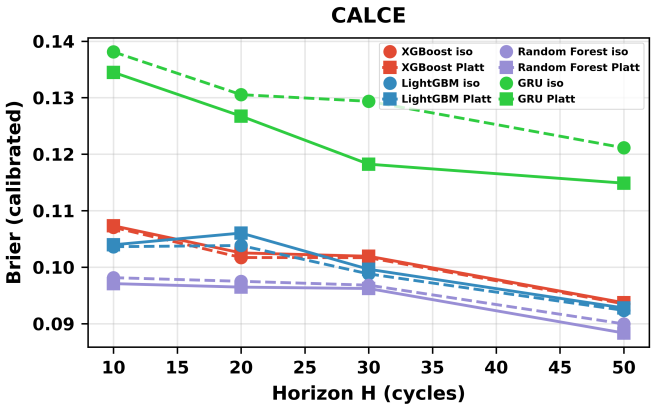


Fig. 4. Calibration curves on CALCE at H=20: raw, Platt, isotonic, and temperature-scaled scores against the observed failure frequency.

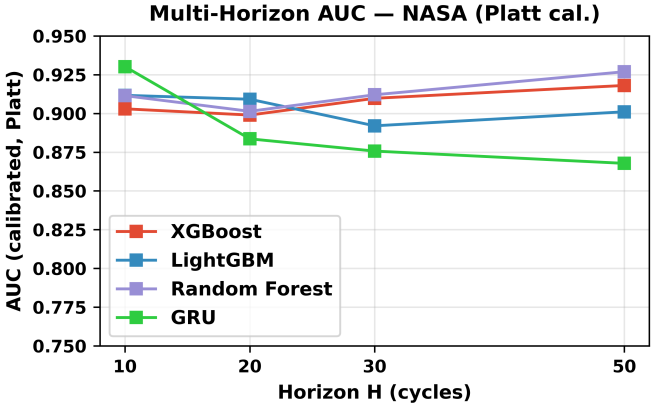


Fig. 5. Multi-horizon pooled AUC for the tree ensembles across $H \in \{10, 20, 30, 50\}$: within-dataset stability against transfer degradation.

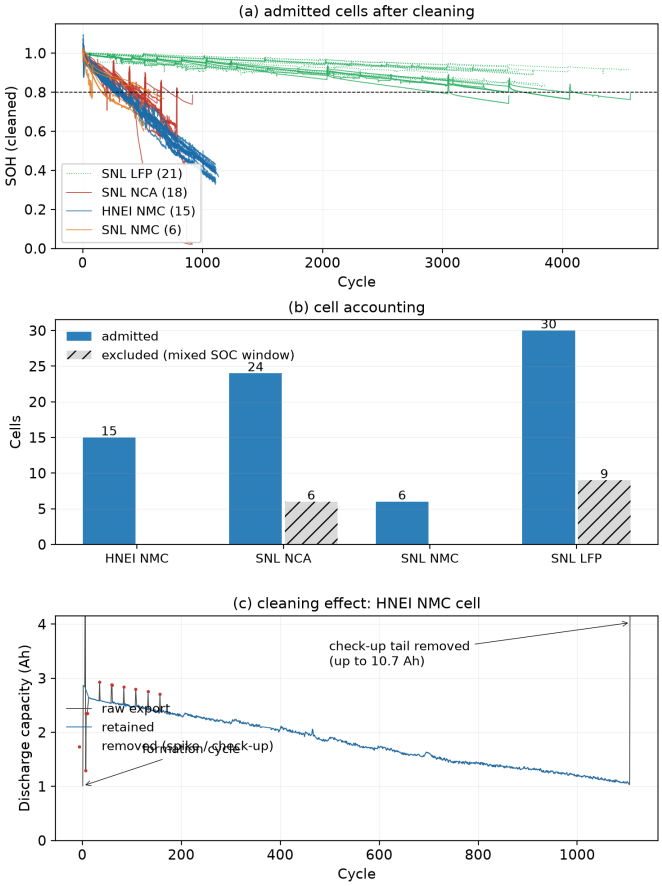


Fig. 6. Data audit of the Battery Archive groups after the pre-registered cleaning rule (formation-cycle drop, Hampel despiking, level-shift truncation, voltage validity, minimum 20 usable cycles): (a) admitted-cell capacity traces with the 80% end-of-life criterion; (b) retained versus excluded cycles per group; (c) recorded exclusion reasons. Of 75 downloaded cells, 60 are admitted and 15 are excluded for exactly one recorded reason.

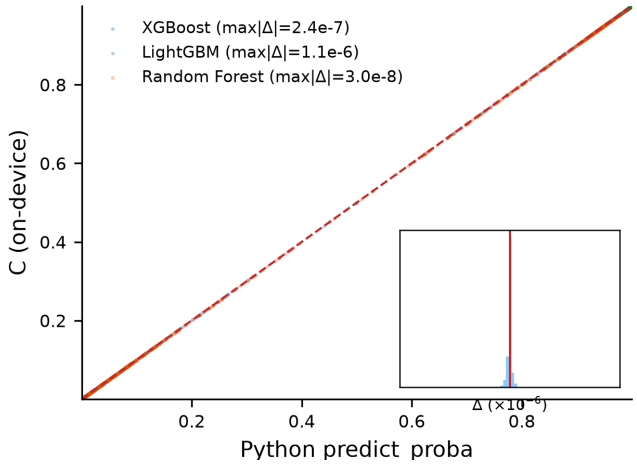


Fig. 7. On-device C engine predictions versus Python `predict_proba()` over all 1 028 rows and three models (agreement scatter on the identity line, inset residual histogram of C minus Python in units of 10^{-6}). Maximum absolute errors are 2.4×10^{-7} (XGBoost), 1.1×10^{-6} (LightGBM), and 3.0×10^{-8} (Random Forest).

TABLE VI
SOH Ablation Evidence (ALL-LCO Source): Pooled AUC With and Without SOH, Paired Cell-Level Bootstrap 95% CI on the Difference (Primary), and Cycle-Level DeLong p-Value (Secondary; Inflated by Within-Cell Correlation).

Target	Model	H	AUC w/ SOH	AUC w/o	Delta AUC [boot 95% CI]	DeLong p
Oxford	XGBoost	10	0.980	0.433	+0.547 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Oxford	XGBoost	20	0.917	0.429	+0.488 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Oxford	XGBoost	30	0.991	0.377	+0.614 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Oxford	XGBoost	50	0.932	0.580	+0.351 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Oxford	LightGBM	10	0.996	0.547	+0.449 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Oxford	LightGBM	20	0.971	0.332	+0.639 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Oxford	LightGBM	30	0.743	0.484	+0.259 (CI undefined: 5 cells)	$9.7e-13$
Oxford	LightGBM	50	0.832	0.448	+0.384 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Oxford	Random Forest	10	0.978	0.626	+0.352 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Oxford	Random Forest	20	0.989	0.581	+0.409 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Oxford	Random Forest	30	0.971	0.462	+0.508 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Oxford	Random Forest	50	0.933	0.460	+0.473 (CI undefined: 5 cells)	$< 10^{-16}$ (underflow)
Severson	XGBoost	10	0.878	0.741	+0.137 [0.079, 0.249]	$< 10^{-16}$ (underflow)
Severson	XGBoost	20	0.896	0.750	+0.146 [0.097, 0.233]	$< 10^{-16}$ (underflow)
Severson	XGBoost	30	0.906	0.745	+0.162 [0.115, 0.237]	$< 10^{-16}$ (underflow)
Severson	XGBoost	50	0.917	0.736	+0.180 [0.134, 0.250]	$< 10^{-16}$ (underflow)
Severson	LightGBM	10	0.870	0.736	+0.134 [0.105, 0.200]	$< 10^{-16}$ (underflow)
Severson	LightGBM	20	0.882	0.718	+0.164 [0.127, 0.227]	$< 10^{-16}$ (underflow)
Severson	LightGBM	30	0.872	0.771	+0.102 [0.030, 0.230]	$< 10^{-16}$ (underflow)
Severson	LightGBM	50	0.899	0.744	+0.155 [0.112, 0.214]	$< 10^{-16}$ (underflow)
Severson	Random Forest	10	0.903	0.740	+0.163 [0.133, 0.225]	$< 10^{-16}$ (underflow)
Severson	Random Forest	20	0.889	0.746	+0.143 [0.106, 0.226]	$< 10^{-16}$ (underflow)
Severson	Random Forest	30	0.919	0.748	+0.171 [0.138, 0.232]	$< 10^{-16}$ (underflow)
Severson	Random Forest	50	0.921	0.746	+0.175 [0.114, 0.264]	$< 10^{-16}$ (underflow)

TABLE VII
Arm A Recalibration on Severson Without SOH (H=20): Values Average the Three LCO Sources and Available Tree Models; ECE Uses 10 Equal-Width Probability Bins. From the Released Recalibration Study Artifacts.

k	Zero ECE	Iso. ECE	Platt ECE	Temp. ECE	Iso. AUC	Platt AUC	Temp. AUC
5	0.817	0.044	0.045	0.563	0.616	0.599	0.669
10	0.817	0.010	0.010	0.561	0.634	0.646	0.651
20	0.817	0.013	0.013	0.560	0.634	0.655	0.648
40	0.817	0.015	0.015	0.561	0.637	0.657	0.653

TABLE VIII
Arm B Recovery at $k = 5$ on Severson Without SOH, H=20. Recovery Ratio Uses the Within-LCO Ceiling; Retention Is the LCO-Holdout AUC Before and After the Update. *Ratio above 1.0 is a ceiling-mismatch artifact, not super-ceiling recovery (see main manuscript, Section III-H).

Training	Model	Zero AUC	Arm B AUC	Recovery	DeLong p	LCO retention
NASA	XGBoost	0.620	0.854	+1.04*	$< 10^{-3}$	1.000 → 1.000
NASA	LightGBM	0.493	0.838	+0.88	$< 10^{-3}$	1.000 → 0.997
NASA	Random Forest	0.724	0.751	+0.19	$< 10^{-3}$	0.959 → 0.314
CALCE	XGBoost	0.686	0.796	+0.53	$< 10^{-3}$	0.997 → 0.996
ALL LCO	XGBoost	0.760	0.828	+0.47	$< 10^{-3}$	0.690 → 0.567

TABLE IX
Leave-Condition-Out Versus Mixed-Cell Folds at H=20 (XGBoost). Conditions Are Temperature Band, C-Rate, or Both; Metadata Never Enters the Features. Ref. Is the Ordinary Mixed-Cell AUC for the Same Feature Set (— Where Unstable or Not Run).

Group	Held-out axis	With SOH	Ref.	No SOH	Ref.
SNL NCA	temp × rate	0.989	0.985	0.894	0.928
SNL NCA	temperature	0.989	0.985	0.888	0.928
SNL NCA	C-rate	0.961	0.985	0.820	0.928
SNL NMC	temp × rate	0.847	—	0.666	—
SNL NMC	temperature	0.787	—	0.569	—
SNL NMC	C-rate	0.847	—	0.666	—
SNL LFP	temp × rate	0.991	0.994	—	—
SNL LFP	temperature	0.995	0.994	—	—
SNL LFP	C-rate	0.994	0.994	—	—

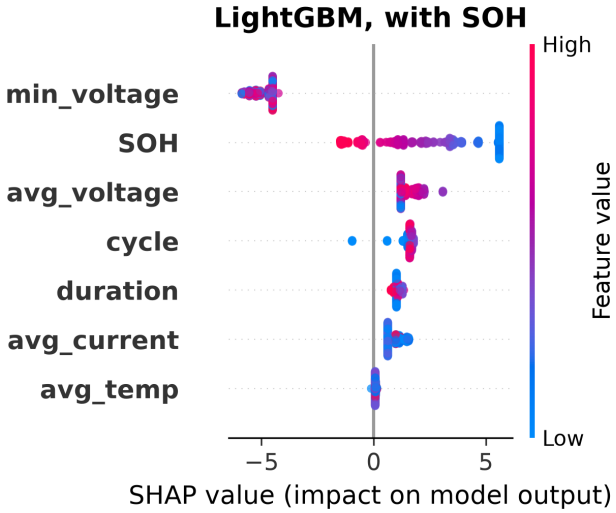


Fig. 8. SHAP feature importance for LightGBM in NASA-to-Oxford transfer at H=20, with SOH. SOH dominates by a wide margin (cf. main manuscript, Fig. 5).

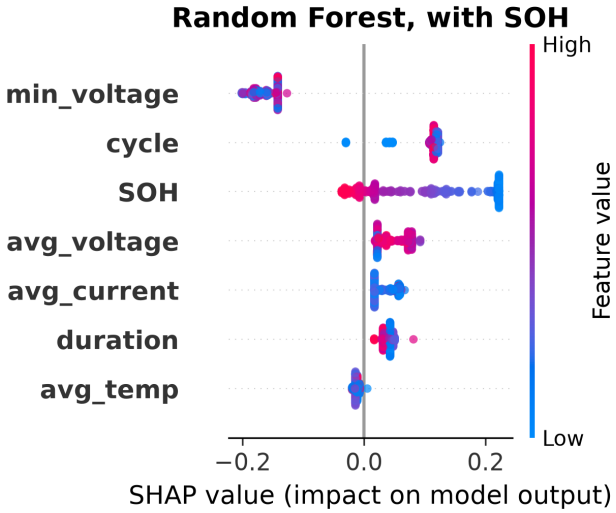


Fig. 9. SHAP feature importance for Random Forest in NASA-to-Oxford transfer at H=20, with SOH. SOH dominates by a wide margin.

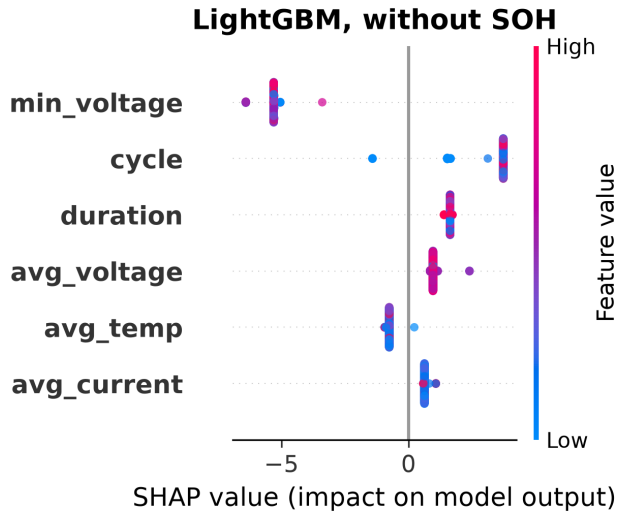


Fig. 10. SHAP feature importance for LightGBM in NASA-to-Oxford transfer at H=20, without SOH. All features collapse to near-zero spread.

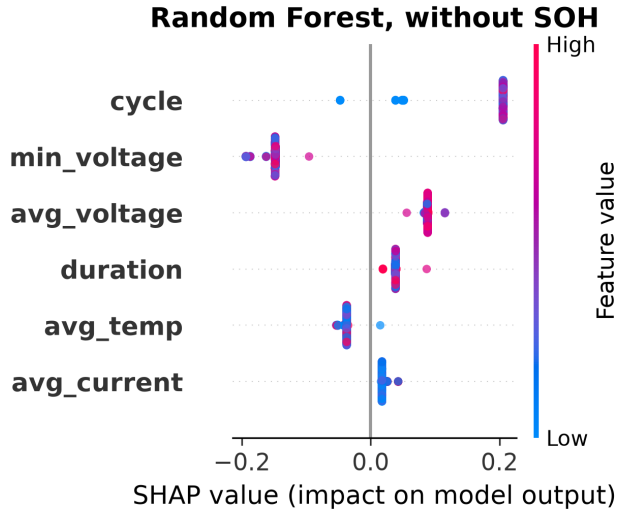


Fig. 11. SHAP feature importance for Random Forest in NASA-to-Oxford transfer at H=20, without SOH. All features collapse to near-zero spread.

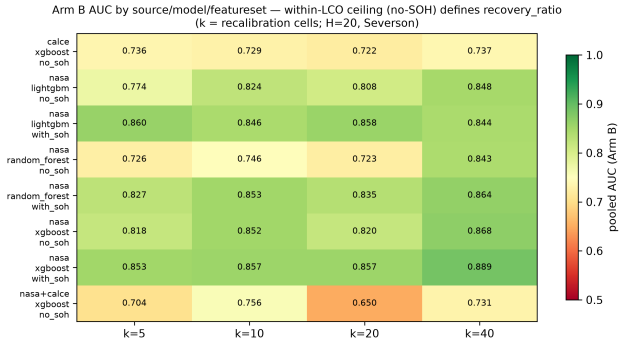


Fig. 12. Recalibration heatmap: calibration error across target-set sizes and calibrators for the Arm A protocol (main manuscript, Section III-H).

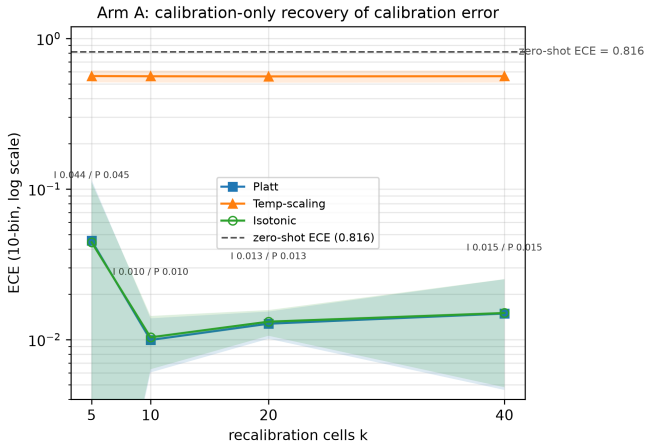


Fig. 13. Arm A recalibration: expected calibration error versus number of labeled target cells k on Severson without SOH.

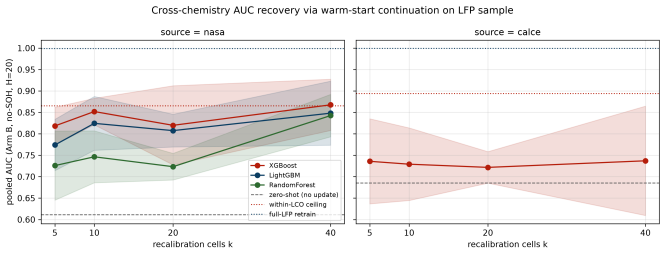


Fig. 14. Arm A recalibration: pooled AUC versus number of labeled target cells k on Severson without SOH — calibration error is repaired; discrimination is not.